# Eduardo Mario Gutiérrez

**Last update: Oct 27, 2026**

**Citizenship: Argentina/Italy**


CONTACT INFORMATION

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PROFESSIONAL SUMMARY

I am a theoretical astrophysicist who uses semi-analytical models and large-scale numerical simulations, in particular, general-relativistic Magnetohydrodynamic (GRMHD) simulations, to study some of the extreme phenomena that take place in the vicinity of compact objects. These include accretion flows onto single and binary black holes, relativistic jets, and binary neutron star mergers. I aim to understand how these systems produce high-energy multimessenger radiation, including electromagnetic waves across the spectrum, gravitational waves, and neutrinos.

POSITIONS

**Postdoctoral Researcher** Oct 2026–present
- *Place:* Max Planck Institute for Gravitational Physics (Albert Einstein Institute), Potsdam, Germany

**Postdoctoral Scholar** 2023–2026
- *Place:* Institute for Gravitation and the Cosmos, Penn State University, USA

EDUCATION

**Ph.D. in Astronomy** Oct 2022
FCAG-UNLP, La Plata, Bs. As., Argentina
- *Thesis:* Radiation from accreting black holes: micro- and macrophysical problems. **Mark:** 10.0/10.0
- *Advisor*: Prof. Gustavo E. Romero. *Co-advisor*: Dr. Florencia L. Vieyro

**Licenciado en Astronomía (equiv. to M.Sc. in Astronomy)** 2011–2017
FCAG-UNLP, La Plata, Bs. As., Argentina
- *GPA*: **9.79/10.0**. Ranked first in class.
- *Thesis*: Cosmology and primordial black holes. **Mark**: 10.0/10.0
- *Advisor*: Prof. Gustavo E. Romero. *Co-advisor*: Dr. Florencia L. Vieyro

FELLOWSHIPS AND AWARDS

- CONICET Doctoral Fellowship 2017
- *2016 Distinguished Graduate (Astronomy)*, for the highest mark of the promotion UNLP, 2017
- *Joaquín V. González Award*, for outstanding marks Government of La Plata, 2017

RESEARCH STAYS AND SCHOOLS

- Visitor at the Perimeter Institute (five days) Jun 2026
- Visitor at Columbia University and the CCA, Flatiron Institute (four days) Oct 2024
- Visitor at the Perimeter Institute (a week) Sep 2024
- Visitor at KIPAC, Stanford University (2 days) Apr 2024
- Visitor at the Institut d'Astrophysique de Paris (a week) Oct 2023
- Visitor at the Center for Computational Relativity and Gravitation of the Rochester Institute of Technology (CCRG-RIT) (a week) Jan 2023
- Visiting scholar at the Center for Computational Relativity and Gravitation of the Rochester Institute of Technology (CCRG-RIT) Sep 2020 - Feb 2021
  *Advisor:* Prof. Manuela Campanelli
  *Topic:* Electromagnetic emission from supermassive binary black holes

- School on High Energy Astrophysics — Aug 2019
  *Place:* ICTP-SAIFR, São Paulo, São Paulo, Brasil
- ISAPP 2019, Cosmic Ray Vision from the Southern Sky — Mar 2019
  *Place:* Pierre Auger Observatory, Malargue, Mendoza, Argentina.
- SPSAS-HighAstro, São Paulo School of Advance Science in High Energy Astrophysics in the CTA era — May 2017
  *Place:* IAG-USP, São Paulo, São Paulo, Brasil.
- IFT-Perimeter-SAIFR Journeys into Theoretical Physics — Jun 2016
  *Place:* ICTP-SAIFR, São Paulo, São Paulo, Brasil.

## Mentoring

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|---|---|---|
| 2023- | Tomás Mazzei | advisor of M.Sc. thesis in Astronomy, UNLP. |
| 2023 | Louis Buchalter | co-advisor of REU summer research program, Penn State |
| 2023-2024 | Nilay Mancini | co-advisor of a short research program, Penn State |

## Grants and travel funding

2023- Co-PI in allocation award: "Numerical Relativity Simulations of Compact Binary Mergers". Total Award: 238 million core hours in Frontera (TACC, USA) (PI: David Radice)

2021 Collaborator in NSF grant: "Collaborative Research: Supermassive Binary Black Hole Mergers: Accretion Dynamics and Electromagnetic Output" (at NSF Windows on the Universe: The Era of Multi-messenger Astrophysics) (PI: Manuela Campanelli and Julian Krolik)

2020 Collaborator in allocation award: "Supermassive Black Hole Approaching Merger: Accretion Dynamics, Jets and Electromagnetic Signals". Total Award: 230 million core hours in Frontera (TACC, USA) (PI: Manuela Campanelli)

2016,2017,2018,2019 Full funding given by FAPESP and the ICTP to attend 3 schools at the ICTP-SAIFR and at IAG-USP, São Paulo, Brasil; Full funding given by the Pierre Auger Observatory for the two-week school 'ISAPP 2019: Cosmic Ray Vision from the Southern Sky' in Malargue, Mendoza, Argentina

## Presentations/ Colloquia

- Invited talk, Perimeter Institute Strong Gravity Seminar (ON, Canada) — Jun 2026
- Invited talk, Perimeter Institute Strong Gravity Seminar (ON, Canada) — Sep 2024
- Invited talk, IAR Colloquium (Bs. As., Argentina) — Apr 2024
- Invited talk, FCAGLP Seminar (Bs. As., Argentina) — Apr 2024
- Invited talk, Tea Talk at KIPAC, Stanford (CA, USA) — Apr 2024
- Invited talk, HEPRO VIII (Paris, France) — Oct 2023
- Invited talk, GdR Ondes gravitationnelles workshop (Paris, France) — Oct 2023
- Invited talk, New Frontiers in GRMHD Simulations of Accreting Black Holes (Virtual) — Apr 2023
- Invited talk, CCRG-RIT Lunch meeting (NY, USA) — Apr 2023
- Invited talk, Black Hole Group (CIERA, Northwestern University) meeting (Virtual) — Nov 2021
- Invited talk, Black Hole Group (IAG-USP, Brazil) meeting (Virtual) — Apr 2021
- Invited talk, IAR Colloquium (Virtual) — Apr 2021
- Invited talk, CCRG-RIT Lunch meeting (NY, USA) — Nov 2020
- **13** Contributed talks at conferences, including ICRC2025, GR25-Amaldi16, TeVPA2024, and others.

## Teaching experience

More than **7 years as a teaching assistant and 2 years as an instructor** in several courses of mathematics, physics, and astrophysics at the Universidad Nacional de La Plata. Including: *Black Hole Astrophysics, Relativistic Astrophysics, Calculus, Electromagnetism, and Classical Mechanics.*

## Computational Skills

- *Programming Languages*: C/C++, Python, Fortran, LaTeX.
- *Astrophysical Software*: `Athena++`, `AthenaK` (Kokkos), `Harm3D`, `Bothros` (GRRT)

- *Use of HPC infrastructure:* Frontera (TACC), NERSC clusters, Aurora (ALCF)

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**Outreach activities**

Various outreach talks and articles. *Detail:*

- Articles: The jet of Centaurus A as never seen before (IAR Bulletin, Sep 2021), *Multimessenger astronomy* (IAR Bulletin, Sep 2019), The first image of a black hole (IAR Bulletin, Jun 2019)
- Public talks: *The first image of a black hole* (La Plata Planetarium, May 2019), *What is a black hole?*

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**Professional service**

- Reviewer in scientific journals and institutions: The Astrophysical Journal Letters, The Astrophysical Journal, Astronomy & Astrophysics
- Co-organizer, 1st Latin American Meeting of Scientific Philosophy, Bs. As., Argentina — Sep 2015

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**Refereed Publications**

**11** as first author, **4** as second/third author; **23** in total. For the full list, go to my **ADS Library**.

1. **Gutiérrez, E.M.**; del Palacio, S.; Mazzei, T.T.
   *The accretion flow and jet of M87: broadband modelling across spatial scales*
   arXiv:2610.26434 (submitted, 2026)

2. Zhu, H.; Dittmann, A.J.; Zhang, L.; Stone, J.M.; **Gutiérrez, E.M.**; Radice, D.
   *Finite-Solid-Angle Boltzmann Radiation Transport on Dynamical Spacetimes in AthenaK*
   arXiv:2608.22020 (2026)

3. Cook, W.; **Gutiérrez, E.M.**; Bernuzzi, S.; Radice, D.; Daszuta, B.; Fields, J.; Hammond, P.; Bandyopadhyay, H.; Jacobi, M.
   *Magnetic Field Configurations in Binary Neutron Star Mergers II: Inspiral, Merger and Ejecta*
   Phys. Rev. D, 114, 063050 (2026)

4. **Gutiérrez, E.M.**; Cook, W.; Radice, D.; Bernuzzi, S.; Fields, J.; Hammond, P.; Daszuta, B.; Bandyopadhyay, H.; Jacobi, M.
   *Magnetic Field Configurations in Binary Neutron Star Mergers I: Post-merger Remnant and Disk*
   Phys. Rev. D, 114, 063049 (2026)

5. Daszuta, B.; Bernuzzi, S.; Jacobi, M.; **Gutiérrez, E.M.**; Hammond, P.; Cook, W.; Radice, D.
   *Binary neutron star merger simulations with neutrino transport using GR-Athena++*
   Phys. Rev. D, 114, 044062 (2026)

6. **Gutiérrez, E.M.**; Radice, D.; Fields, J.; Stone, J.M.
   *Turbulent Dynamo Action in Binary Neutron Star Mergers*
   arXiv:2601.20953 (**accepted** in Phys. Rev. Lett., 2026)

7. Daszuta, B.; Cook, W.; Hammond, P.; Fields, J.; **Gutiérrez, E.M.**; Bernuzzi, S.; Radice, D.
   *Numerical relativity simulations of compact binaries: comparison of cell- and vertex-centered adaptive meshes*
   Phys. Rev. D, 112, 103006 (2025)

8. **Gutiérrez, E.M.**; Bhattacharya, M.; Radice, D.; Murase, K.; Bernuzzi, S.
   *Cocoon shock breakout emission from binary neutron star mergers*
   Phys. Rev. D, 111, 063031 (2025)

9. Porter, K.; Noble, S.C.; **Gutiérrez, E.M.**; Pelle, J.; Campanelli, M.; Schnittman, J.; Kelly, B.
   *A Parameter Study of the Electromagnetic Signatures of an Analytical Mini-Disk Model for Supermassive Binary Black Hole Systems*
   ApJ, 979, 2 (2025)

10. **Gutiérrez, E.M.**; Combi, L.; Ryan, G.
*Accretion onto supermassive black hole binaries*
Invited chapter in *"New Frontiers in GRMHD Simulations"* (Eds. C. Bambi, Y. Mizuno, S. Shashank, and F. Yuan), Springer Singapore, pp. 447–492 (2025); DOI: 10.1007/978-981-97-8522-3_14

11. **Gutiérrez, E.M.**; Combi, L.; Romero, G.E.; Campanelli, M.
*Non-thermal radiation from dual jet interactions in supermassive black hole binaries*
MNRAS, 532, 1, 2024

12. Combi, L.; Yang, H.; **Gutiérrez, E.M.**; Noble, S.C.; Romero, G.E.; Campanelli, M.
*General relativistic magnetohydrodynamical simulations of accretion flows through traversable wormholes*
Phys. Rev. D, 109, 103034 (2024)

13. Zubieta, E.; Missel, R.; Sosa Fiscella, V.; Lousto, C.O.; del Palacio, S.; López Armengol, F.G.; García, F.; Combi, J.A.; Wang, L.; Combi, L.; Gancio, G.; Negrelli, C.; **Gutiérrez, E.M.**
*First results of the glitching pulsar monitoring programme at the Argentine Institute of Radioastronomy*
MNRAS, 521, 3 (2023)

14. **Gutiérrez, E.M.**; Combi, L.; Noble, S.C.; Campanelli, M.; Krolik, J.H.; López Armengol, F.G.; García, F.
*Electromagnetic Signatures From Supermassive Binary Black Holes Approaching Merger*
ApJ, 928, 137 (2022)

15. Lousto, C.O.; Missel, R.; Prajapati, H.; Sosa Fiscella, V.; López Armengol, F.G.; Gyawali, P.K.; Wang, L.; Cahill, N.; Combi, L.; del Palacio, S.; Combi, J.A.; Gancio, G.; García, F.; **Gutiérrez, E.M.**; Hauscarriaga, F.
*Vela Pulsar: Single Pulses Analysis with Machine Learning Techniques*
MNRAS, 509, 4 (2021)
*Contribution:* Progamming of some routines and data management.

16. Sosa Fiscella, V.; del Palacio, S.; Combi, L.; Lousto, C.O.; Combi, J.A.; Gancio, G.; García, F.; **Gutiérrez, E.M.**; Hauscarriaga, F.; Kornecki, P.; López Armengol, F.G.; Mancuso, G.C.; Muller, A.L.; Simaz Bunzel, A.
*PSR J0437-4715: The Argentine Institute of Radioastronomy 2019-2020 Observational Campaign*
ApJ, 908, 158 (2021)
*Contribution:* Progamming of some routines and data management.

17. **Gutiérrez, E.M.**; Vieyro, F.L.; Romero G.E.
*Nonthermal processes in hot accretion flows onto supermassive black holes: An inhomogeneous model*
A&A, 649, A87 (2021)

18. Romero, G.E.; **Gutiérrez, E.M.**
*The Origin of Matter at the Base of Relativistic Jets in Active Galactic Nuclei*
Universe, 6(7), 99 (2020)

19. **Gutiérrez, E.M.**; Romero, G.E.; Vieyro, F.L.
*Cosmic rays from the nearby starburst galaxy NGC 253: the effect of a low luminosity active galactic nucleus*
MNRAS, 494, 2 (2020)

20. **Gutiérrez, E.M.**; Nemmen, R.; Cafardo, F.
*A Nonthermal Bomb Explains The Near-infrared Superflare of Sgr A**
ApJL, 891, 2, L36 (2020)

21. Gancio, G.; Lousto, C.O.; Combi, L.; del Palacio, S.; López Armengol, F.G.; Combi, J.A.; García, F.; Kornecki, P.; Muller, A.L.; **Gutiérrez, E.M.**; Hauscarriaga, F.; Mancuso, G.C.
*Upgraded antennas for pulsar observations in the Argentine Institute of Radio astronomy*
A&A, 633, A84 (2020)
*Contribution:* Programming of some routines and data management.

22. Pérez, D.; Romero, G.E.; Combi, L.; **Gutiérrez, E.M.**
*A note on geodesics in inhomogeneous expanding spacetimes*
Class. Quantum Grav., 36, 5 (2019)

23. **Gutiérrez, E.M.**; Vieyro, F.L.; Romero, G.E.
*Primordial black hole evolution in two-fluid cosmology*
MNRAS, 473, 4 (2018)

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CONFERENCE PROCEEDINGS

- **Gutiérrez et al.**, Radiation from hot accretion flows, BAAA, Vol. 61 (2019)
- **Gutiérrez et al.**, A gas of cosmological black holes, BAAA, Vol. 60 (2018)

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PUBLICATIONS IN PREP.

- **Gutiérrez et al.**, Kinematic and dynamic electromagnetic signatures in close supermassive black hole binaries

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